# Is rooftop solar in France reliably low-carbon?

Uncertainty and sensitivity analysis of a 7-parameter life cycle model of rooftop PV electricity

**28 g**
$CO_2$eq per kWh, median of 1,000 Monte Carlo runs

**17–45 g**
90% of runs fall in this range

**12.3%**
chance of exceeding the 40 g threshold

**30%**
of the variance comes from solar irradiation (Sobol')

**−25%**
lower footprint with modules made on EU electricity

> Usually, but not everywhere. One run in eight exceeds 40 g $CO_2$eq/kWh, mostly at low-sunlight sites using energy-intensive silicon. Local irradiation and lifetime data would remove the largest share of the uncertainty.

Brightway · Excel · Monte Carlo (1,000 runs) · Elasticities & tornado · Spearman · Sobol' indices · Morris screening · Factor mapping · Paired comparison · GWP100

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# One equation, seven uncertain inputs, and a reference case of 27.4 g $CO_2$eq/kWh

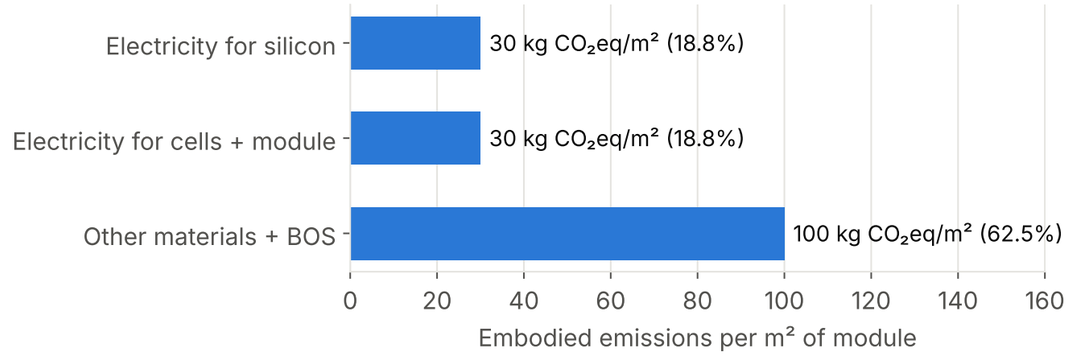
**Functional unit:** 1 kWh of AC electricity from a rooftop multi-Si PV system installed in France. **Impact:** climate change, GWP100.

```
Y = 1000 × [ CI_mix × (m_Si × elec_Si + E_cm) + M ]
    / [ η × G × PR × LT × (1 − d × (LT − 1) / 2) ]
```

| Input | Meaning | Reference | Distribution |
|---|---|---|---|
| **η** | Module efficiency | 0.175 | Triangular (0.15, 0.175, 0.22) |
| **G** | In-plane irradiation | 1,500 kWh/m²/yr | Uniform (1,100–1,900), north to south France |
| **PR** | Performance ratio | 0.80 | Uniform (0.75–0.85) |
| **LT** | Lifetime | 30 yr | Normal (30, sd 4) |
| **elec_Si** | Electricity for silicon | 30 kWh/kg Si | Triangular (11, 30, 110), FBR to Siemens process |
| **CI_mix** | Manufacturing grid intensity | 1.0 kg $CO_2$eq/kWh | Country by market share: China 71%, Japan 16%, Korea 7%, Malaysia 6% |
| **M** | Other materials + balance of system | 100 kg $CO_2$eq/m² | Uniform (60–140) |

Fixed: silicon 1 kg/m², cell + module electricity 30 kWh/m², degradation 0.5%/yr.

CONTRIBUTION ANALYSIS AT THE REFERENCE CASE

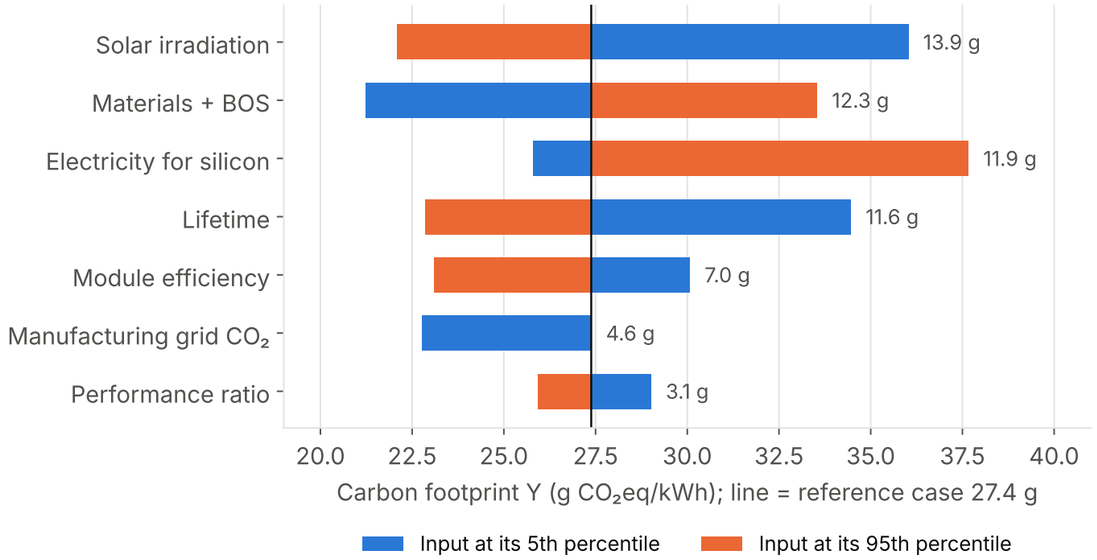



Materials and balance of system make up **62.5%** of embodied emissions. The two electricity steps each add 18.8%.

But a contribution analysis only covers the numerator. Efficiency, irradiation, performance ratio and lifetime set the electricity produced (5,843 kWh/m² at the reference), and they never appear in it. That is why the study needs a sensitivity analysis.

# Same elasticity, very different impact: the range of each input is what matters

TORNADO: EACH INPUT MOVED TO ITS 5TH AND 95TH PERCENTILE




**Elasticities** (+1% on one input): efficiency, irradiation and performance ratio all give **−0.99**. A 1% rise in any of them cuts the footprint by about 1%. Lifetime gives −0.91, materials +0.63, grid intensity +0.38, silicon electricity +0.19.

**The tornado separates them.** Irradiation moves the result from 22.1 to 36.0 g across its range; performance ratio only from 25.9 to 29.0 g. Same local slope, very different real-world spread.

FIRST-ORDER SHARE OF VARIANCE (ELASTICITY × RELATIVE SPREAD)

| Input | Share |
|---|---|
| Solar irradiation | 26.3% |
| Materials + BOS | 23.5% |
| Electricity for silicon | 20.3% |
| Lifetime | 16.6% |
| Module efficiency | 7.6% |
| Manufacturing grid | 4.2% |
| Performance ratio | 1.4% |

Performance ratio could be fixed at 0.80 without losing much.

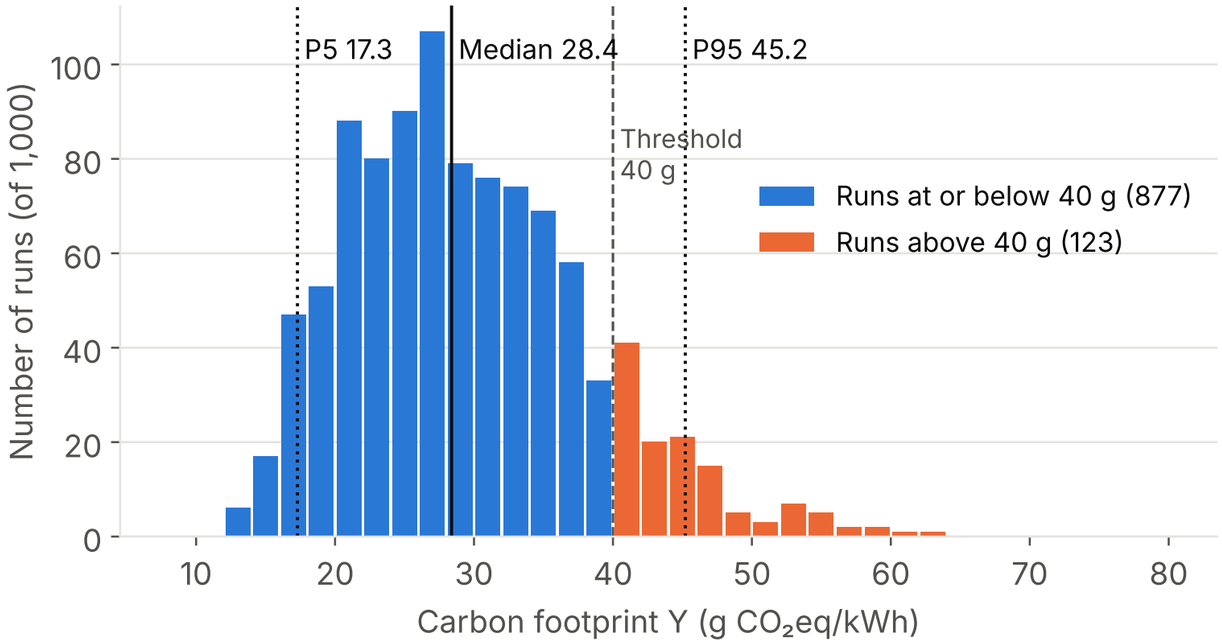
# 1,000 Monte Carlo runs: a median of 28 g, but one run in eight exceeds 40 g

**28.4 g**
median footprint

**29.5 g**
mean (pulled up by the right tail)

**17.3–45.2 g**
90% interval (P5–P95)

**12.3%**
P(Y > 40 g): 123 of 1,000 runs



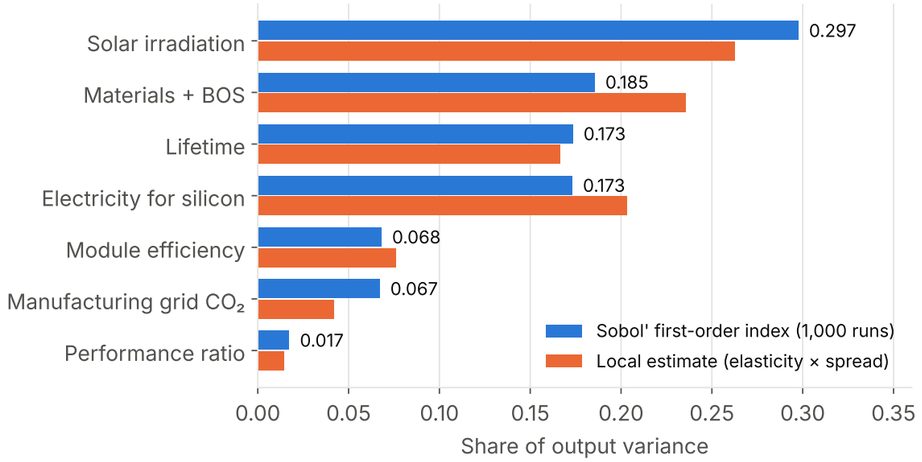

The reference case (27.4 g) sits below both the median and the mean. A few high-footprint runs form a long right tail and pull the mean up, so I report the **median with the 90% range and the probability of exceeding the threshold**, and show the mean alongside.

Inputs were sampled with inverse CDFs from 1,000 fixed random numbers (seed 2026), so every result here can be reproduced exactly.

**What this does not tell us:** which inputs cause the spread, how they interact, or why some runs cross the line. That is the job of the global sensitivity analysis.

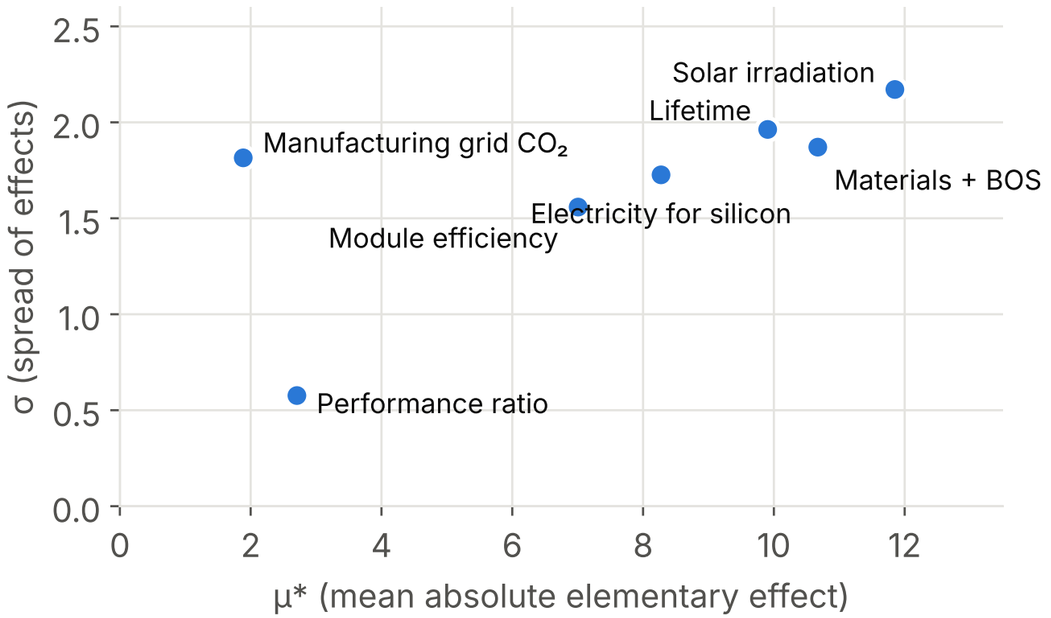
# Irradiation drives the uncertainty, and three methods agree on the top four inputs

## SOBOL' FIRST-ORDER INDICES VS THE LOCAL ESTIMATE




| Input | Sobol' $S_i$ | Spearman ρ | Morris μ* |
|---|---|---|---|
| Solar irradiation | 0.297 | -0.55 | 11.85 |
| Materials + BOS | 0.185 | +0.44 | 10.67 |
| Lifetime | 0.173 | -0.41 | 9.91 |
| Electricity for silicon | 0.173 | +0.39 | 8.27 |
| Module efficiency | 0.068 | -0.26 | 7.01 |
| Manufacturing grid | 0.067 | +0.26 | 1.89 |
| Performance ratio | 0.017 | -0.12 | 2.71 |

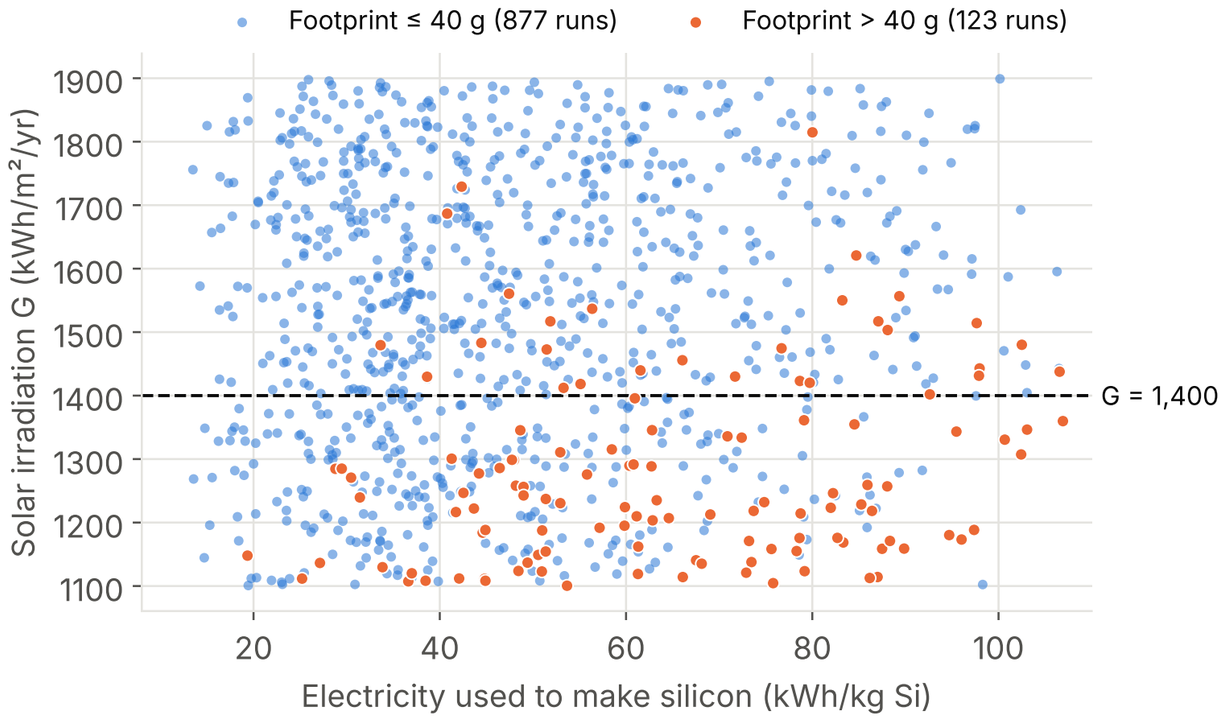
## MORRIS SCREENING: 80 MODEL RUNS (10 TRAJECTORIES)




With 80 runs instead of 1,000, Morris already ranks irradiation, materials, lifetime and silicon electricity at the top, like Sobol'. It screens inputs cheaply but does not give their share of variance.

The Sobol' indices sum to **0.981**, so the inputs act almost independently. The local estimate overweights silicon electricity (20% vs 17%) because one slope cannot capture its wide, skewed range. The grid input's σ close to μ* comes from sampling a four-country input on a coarse grid, not from a physical effect.

# When does PV fail the 40 g test? Low sunlight combined with energy-intensive silicon




Above the dashed line (irradiation ≥ 1,400 kWh/m²/yr) only 29 of 643 runs fail (4.5%). China supplies 87% of failing runs against 69% of passing ones. Points overlap because the other five inputs also move the result.

## FAILING RUNS VS PASSING RUNS (DIFFERENCE IN MEAN INPUT)

| | |
|---|---|
| Electricity for silicon | +36% |
| Materials + BOS | +17% |
| Manufacturing grid | +8% |
| Module efficiency | -3% |
| Performance ratio | -1% |
| Lifetime | -10% |
| Solar irradiation | -16% |

## HOW THE ASSUMPTIONS CHANGE THE ANSWER

| Scenario | S_LT | P(Y > T) |
|---|---|---|
| Lifetime Normal(30,4), T = 40 | 0.173 | **12.3%** |
| Lifetime Uniform(15,40), T = 40 | 0.478 | **26.3%** |
| Lifetime Normal(30,4), T = 30 (RTE production mix) | 0.173 | **43.3%** |
| Lifetime Normal(30,4), T = 52 (ADEME consumption mix) | 0.173 | **1.8%** |

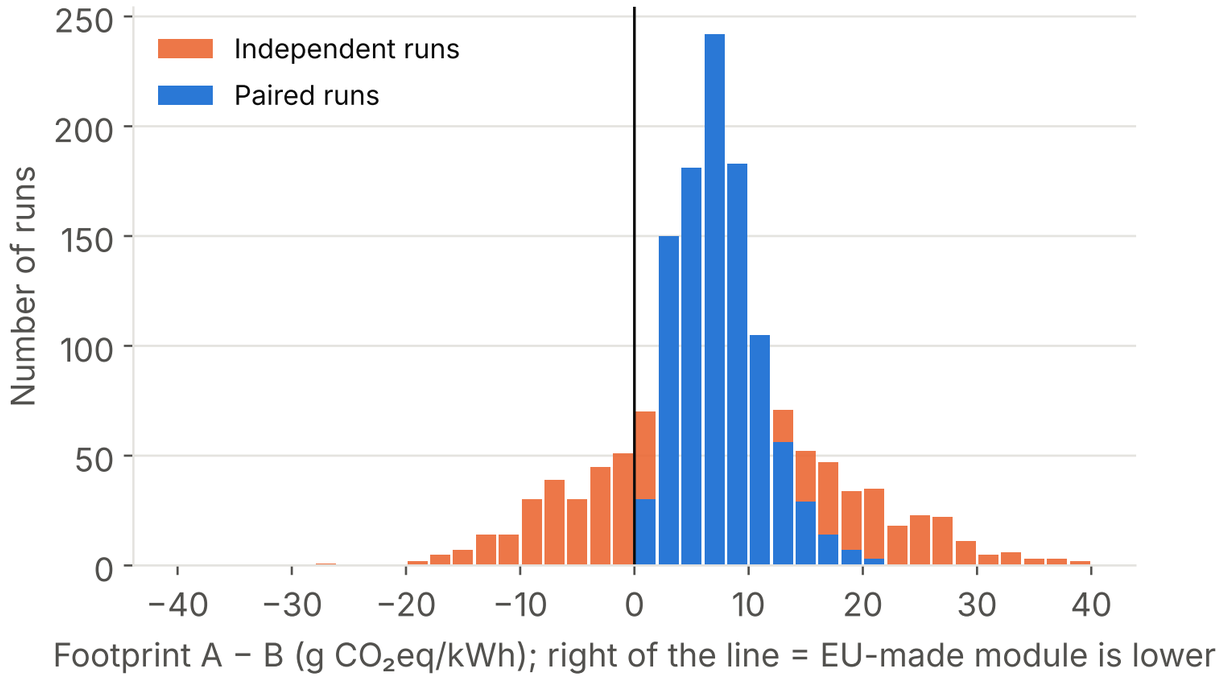
Widening the lifetime assumption doubles the risk and makes lifetime the largest source of variance. For electricity used on site, the consumption-based French mix (≈52 g, ADEME, incl. imports and losses) is the fairer benchmark; the production-based 30 g (RTE) is stricter.

# Modules made on European electricity are lower in every paired run

**29.5 → 22.0 g**
mean footprint: market average (A) → EU-made (B), −25%

**100%**
of paired runs favour B (sd of difference 3.6 g)

**76.2%**
if the runs were not paired (sd 10.7 g)




Option B is the same system on the same roof, with silicon, cells and modules made on European electricity (0.35 kg $CO_2$eq/kWh). Only the manufacturing grid changes.

**Pairing matters.** Irradiation, lifetime and materials move together in A and B, so their uncertainty cancels in the difference. Comparing two separate Monte Carlo runs would make B look better only 76% of the time and triple the spread.

For a comparative LCA, report the paired difference, not two separate ranges.

# Recommendation to the regional energy agency

1. **Do not promote every PV site automatically.** At a 40 g threshold, 12.3% of cases exceed it.
2. **Prioritise sunny sites.** With irradiation of at least 1,400 kWh/m²/yr, only 29 of 643 runs (4.5%) exceed 40 g.
3. **Be explicit about assumptions.** The result depends on the lifetime assumption and on which grid value is used as the benchmark.
4. **Collect data in this order:** local irradiation, realistic lifetimes, then material and silicon-electricity data for the modules actually bought.
5. **Set an acceptable failure rate** and check borderline sites before promoting them.

## WHAT THIS STUDY SHOWS I CAN DO

Parameterised LCA model | Brightway | Monte Carlo propagation | Global sensitivity (Sobol', Morris) | Factor mapping | Comparative LCA, paired sampling | Writing for decision makers

## ABOUT THIS STUDY

Course case study, Advanced Life Cycle Assessment (Session 03), Grenoble INP–UGA, 2026. The model structure and input ranges were provided by the course and are illustrative orders of magnitude for teaching. Analysis, results and recommendation by Muhammad Anas Raza. Model run in Brightway with input data in Excel; figures regenerated in Python from the same 1,000-run sample (seed 2026).